Strategies for the identification of binders suitable for CDKL5 localization and quantification

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SCAN ME

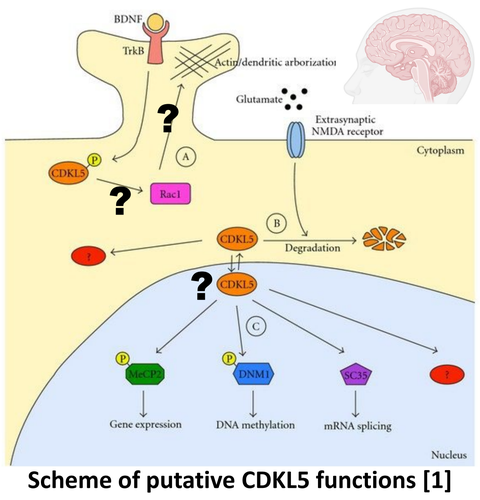
SMASH machine learning for science and humanities postdoctoral program

I FEEL SLOVENIA

Co-funded by the European Union

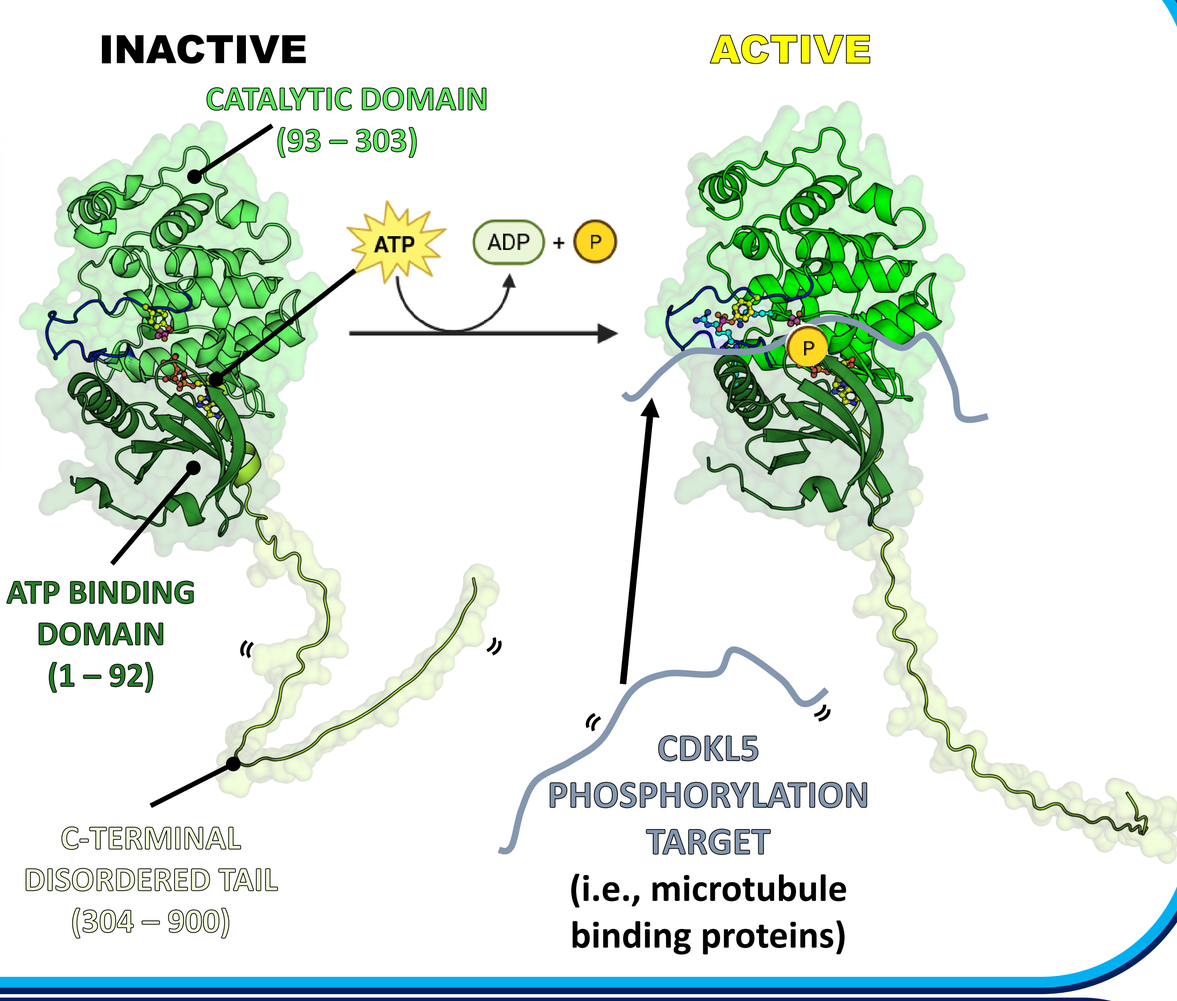
BACKGROUND ON CDKL5

CDKL5 (cyclin-dependent kinase-like 5, Uniprot ID: O76039) is an enzyme that activates upon auto-phosphorylation and then mediates the phosphorylation of microtubule binding proteins and transcription factors essential for crucial brain development [1]. Putative alternative splicing isoforms might exist, with a specific yet unknown region-temporal expression pattern, while non-synonymous genetic mutations can cause CDKL5 Deficiency Disorder, leading to severe neurodevelopmental disabilities from early infancy [2]. To advance our understanding of CDKL5 biology in normal and disease conditions, it is necessary to develop reagents to *in vivo* visualize and quantify the relative amounts of the different isoforms. Traditional IgG antibodies, typically used for specific molecular recognition, have proven difficult to be obtained due to the high sequence identity among CDKL proteins and low immunogenicity of the phylogenetically conserved N-terminal kinase catalytic domain.

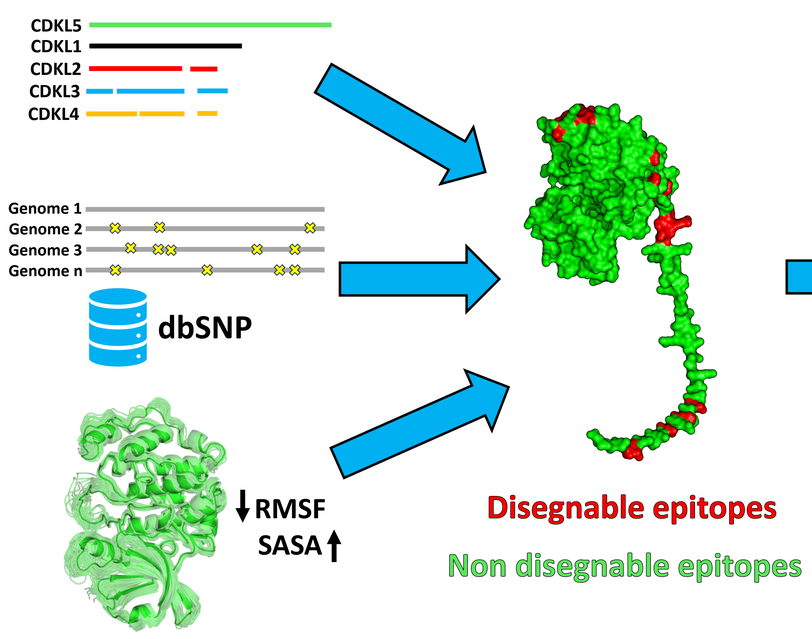


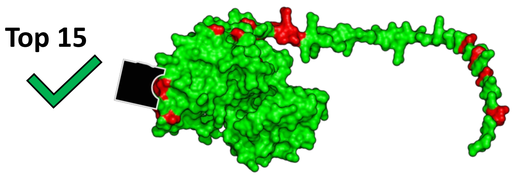
Scheme of putative CDKL5 functions [1]

The C-terminal tail of CDKL5 can be used to develop antibodies, but there is the need to get a larger coverage of the target epitopes for differentiating among the isoforms. Machine Learning *de novo* design methods are promising for making protein binders selective for specific antigen epitopes [3]. Indeed, such methods may represent one of the few available options when working with poorly immunogenic target molecules, unfolded epitopes and highly conserved antigens. Anyway, despite the great acceleration in developing new methods in the last couple of years [4-5], >> 10 designs have still to be characterized to obtain binders with low nM affinity and decent to good developability properties (stable, soluble, non immunogenic and specific for your target). Indeed, CDKL5 has poor solubility for setting a high-throughput *in vitro* screening. During this project, a multi-step workflow has been set for finding ready-to-apply CDKL5 binders, while minimizing the experimental validation work.



HIGH THROUGHPUT *IN SILICO DE NOVO* BINDER DESIGN + SCORING



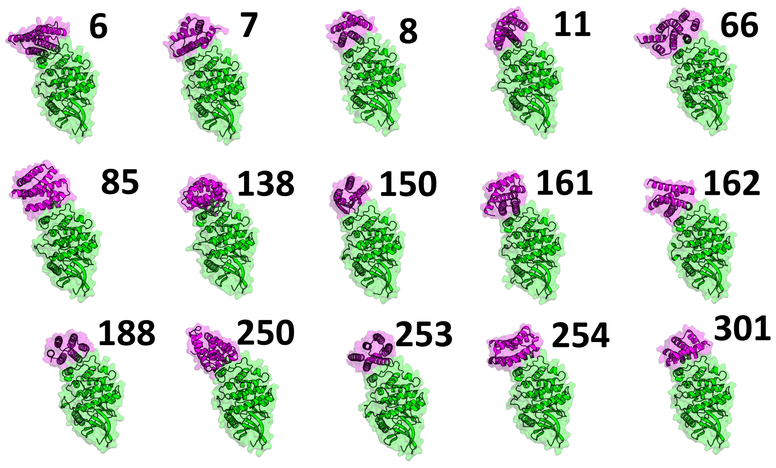


Monomer refolding filters
- Av. plddt > 90 & min > 70 (ESMFold)
- pTM > 0.9 (ESMFold)

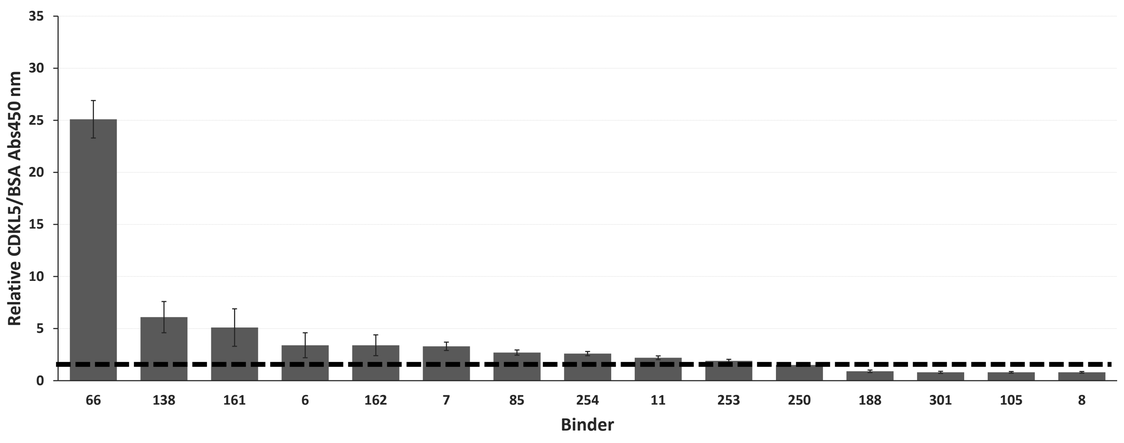
Complex refolding filters
- RosettaDDG/ΔSASA < -20
- Deep learning co-folding < 2 Å
- Rigid-body re-docking < 2 Å
- AI-enhanced MD-simulations binder RMSD with frames superposed along target structure

Selection criteria
Top-15 ipSAE (Boltz-2/AF3)
Structural diversity

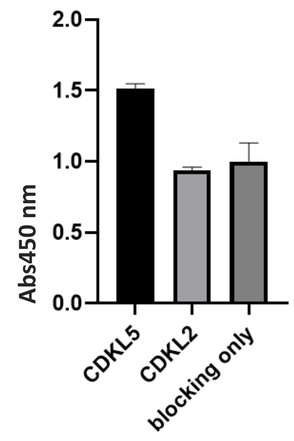
TOP 15 DESIGNS



ELISA ASSAY AT 2 µM

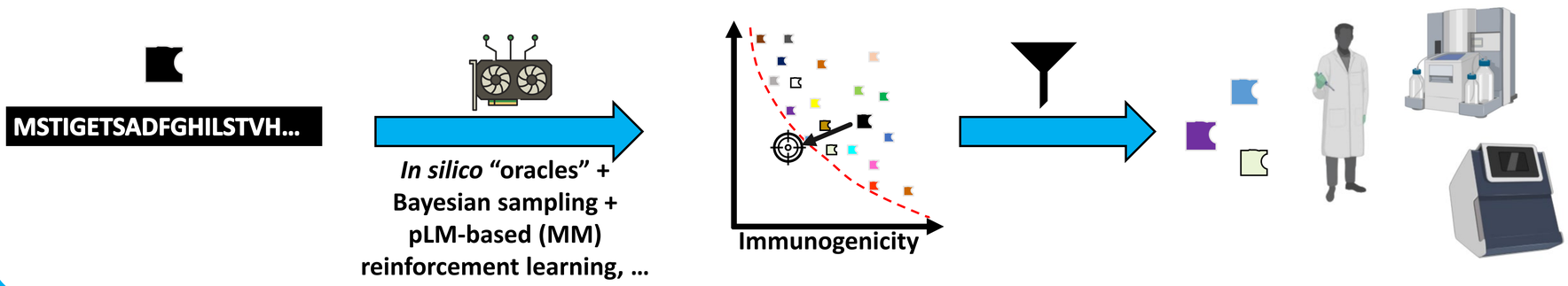


ELISA ASSAY AT 100 nM 66.Fc



66 is the best candidate and specific if assayed against CDKL2 (closest CDKL5 homologue). Now we are functionalizing it

FUTURE GOAL: MULTI-MODAL (MM) LEAD OPTIMIZATION OF BEST DESIGN(S)



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